

Crypto Assign 3

1. Determine the mul inverse modulo 26 for each
1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25 mod 26

Ans i) Mul inverse of 1

$$26 = 1(26) + 0$$

$$1(1) = 1$$

$$1^{-1} = 1 \pmod{26}$$

$$\text{Mul inverse of } 1 = 1$$

ii) 3

$$26 = 3(8) + 2$$

$$3 = 2(1) + 1$$

$$1 = 3 - 2 \quad \text{--- (1)}$$

$$2 = 26 - 3(8)$$

$$1 = 3(a) + 26(-1)$$

$$3^{-1} = a \pmod{26}$$

$$\text{Mul inverse } 3 = a$$

iii) 5

$$26 = 5(5) + 1$$

$$1 = 26 - 5(5)$$

$$1 = 5(5) + 26(-1)$$

$$-5 + 26 = 21$$

$$5^{-1} = 21 \pmod{26}$$

$$5 = 21$$

iv) 7

$$26 = 7(3) + 5$$

$$7 = 5(1) + 2$$

$$5 = 2(2) + 1$$

$$1 = 5 - 2(2) \quad \text{--- (1)}$$

$$2 = 5 - 2(7 - 5)$$

$$1 = 5 - 2 \times 7 + 2 \times 5$$

$$1 = 3(5) - 2(7)$$

$$1 = 3(26 - 7(3)) - 2(7)$$

$$1 = 3 \times 26 - 3 \times (7(3)) - 2(7)$$

$$1 = 3 \times 26 - 11 \times 7$$

$$1 = (-11) \times 7 - 3 \times 26, \quad -11 \times 26 = 15$$

$$7^{-1} = 15 \pmod{26}$$

$$7 = 15$$

v) Mul inverse of 9

$$26 = 9(2) + 8$$

$$9 = 8(1) + 1$$

$$1 = 9 - 8, \quad 8 = 26 - 9(2)$$

$$1 = 9 - (26 - 9(2))$$

$$1 = 9 + 9(2) - 26$$

$$1 = 3(a) - 26$$

$$a^{-1} = 3 \pmod{26}$$

$$9 = 3$$

vi)

vi)

11

$$26 = 11(2) + 4$$

$$11 = 4(2) + 3$$

$$4 = 3(1) + 1$$

$$1 = 4 - 3$$

$$1 = 4 - (11 - 4(2)) = 4 - 11 + 4(2)$$

$$1 = 3(4) - 11$$

$$1 = 3(26 - 11(2)) - 11$$

$$1 = 3 \times 26 - 3 \times (11(2)) - 11$$

$$1 = 3 \times 26 - 7 \times 11$$

$$1 = (-7)11 + 3 \times 26, -7 + 26 \equiv 19$$

$$11^{-1} = 19 \pmod{26}$$

$$11 = 19$$

vii)

15

$$26 = 15(1) + 11$$

$$15 = 11(1) + 4$$

$$11 = 4(2) + 3$$

$$4 = 3(1) + 1$$

$$1 = 4 - 3$$

$$1 = 4 - (15 - 11(1))$$

$$1 = 3(4) - 15$$

$$1 = 3(15 - 11(1)) - 15$$

$$1 = 3 \times 15 - 4 \times 11$$

$$1 = 3 \times 15 - 4 \times 11$$

$$1 = 7(15) - 4(26)$$

$$15^{-1} = 7 \pmod{26}$$

$$15 = 7$$

viii)

17

$$26 = 17(1) + 9$$

$$17 = 9(1) + 8$$

$$9 = 8(1) + 1$$

$$1 = 9 - 8$$

$$1 = 9 - (17 - 9(1))$$

$$1 = 2(9) - 17$$

$$1 = 2(26 - 17(1)) - 17$$

$$1 = 2 \times 26 - 2 \times (17(1)) - 17$$

$$1 = 2 \times 26 - 13 \times 17(1)$$

$$1 = 2 \times 26 - 13 \times 17$$

$$17 = 23$$

ix) 19

$$26 = 19(1) + 7$$

$$19 = 7(2) + 5$$

$$7 = 5(1) + 2$$

$$5 = 2(2) + 1$$

$$1 = 5 - 2(2)$$

$$1 = 5 - 2(7 - 5(1))$$

$$1 = 5 - 2 \times 7 + 2(5(1))$$

$$1 = 3(5) - 2(7)$$

$$1 = 3(19 - 7(2)) - 2(7)$$

$$1 = 3 \times 19 - 8(7)$$

$$1 = 3(19) - 8(26 - 19)$$

$$1 = 11(19) - 8(26)$$

$$19^{-1} = 11 \pmod{26}$$

$$19 = 11$$

x) 21

$$26 = 21(1) + 5$$

$$21 = 5(4) + 1$$

$$1 = 21 - 5(4)$$

$$1 = 21 - 4(26 - 21)$$

$$1 = 21 - 4(26) + 4(21)$$

$$1 = 5(21) - 4(26)$$

$$21^{-1} = 5 \pmod{26}$$

$$21 = 5$$

x1) 23

$$26 = 23(1) + 3$$

$$23 = 3(7) + 2$$

$$3 = 2(1) + 1$$

$$1 = 3 - 2(1)$$

$$1 = 3 - (23 - 2(7))$$

$$1 = 8(7) - 23$$

$$1 = 8(26 - 23) - 23$$

$$1 = (-9)23 + 26(8)$$

$$26 - 9 = 17$$

$$23 = 17$$

xii) 25

$$26 = 25(1) + 1$$

$$1 = 26 - 25$$

$$1 = (7)25 + 26$$

$$26 - 1 = 25$$

$$25^{-1} = 25 \pmod{26}$$

Inv of 25 is 25

2) $P (K_1 \cdot P + K_2) \pmod{26}$

Given $K_1 = 5$, $K_2 = 8$

P-T		$K_1 \cdot P$	$+K_2$		$\pmod{26}$	
G	6	$5 \times 6 = 30$	$30 + 8$	38	12	m
O	14	$5 \times 14 = 70$	$70 + 8$	78	0	a
O	14	$5 \times 14 = 70$	$70 + 8$	78	0	a
D	3	$5 \times 3 = 15$	$15 + 8$	23	23	x
L	11	$5 \times 11 = 55$	$55 + 8$	63	11	L
V	20	$5 \times 20 = 100$	$100 + 8$	108	4	e
C	2	$5 \times 2 = 10$	$10 + 8$	18	18	s
K	10	$5 \times 10 = 50$	$50 + 8$	58	6	g

$$CT = M \times A \times L \times E \times S \times g$$

3.

$$E(K_1 \times P + K_2) \bmod 26$$

$$K_1 = 11, K_2 = 5$$

PT			$+K_2$	$\bmod 26$	Cipher
H	7	$11 \times 7 = 77$	$77 + 5 = 82$	4	E
S	8	$11 \times 8 = 88$	$88 + 5 = 93$	15	P

CT :- EP

4)

$$K^{-1}(C - K_2) \bmod 26$$

$$K_1 = 9, K_2 = 2$$

$$a^{-1} \bmod 26$$

$$26 = 9(2) + 8; 8 = 26 - 9(2)$$

$$9 = 8(1) + 1$$

$$1 = 9 - 8$$

$$1 = 9 - (26 - 9(2))$$

$$1 = 3(9) - 26$$

$$a^{-1} \bmod 26 = 3$$

CT

$$K^{-1}(C - K_2)$$

$$U \quad 20 \quad 3(20 - 2) = 54 \quad C$$

$$C \quad 2 \quad 3(2 - 2) = 0 \quad a$$

$$R \quad 17 \quad 3(17 - 2) = 45 \quad T$$

PT = CAT

5.

$$P = K^{-1}(C - K_2) \bmod 26$$

$$CT = KDYXJUWDEXZY$$

$$K_1 = 3, K_2 = 4$$

$$3^{-1} \bmod 26$$

$$26 = 3(8) + 2, 2 = 26 - 3(8)$$

$$3 = 2(1) + 1$$

$$1 = 3 - 2$$

$$1 = 3 - (26 - 3(8))$$

$$1 = 3(9) - 26$$

$$3^{-1} \bmod 26 = 9$$

CT		$K^{-1}(C-K_1) \pmod{26}$	$\pmod{26}$	PT
K	10	$9(10-4) = 54$	2	C
D	3	$9(3-4) = -9$	17	R
Y	24	$9(24-4) = 180$	24	Y
X	23	$9(23-4) = 171$	15	P
J	9	$9(9-4) = 45$	19	T
V	20	$9(20-4) = 144$	14	O
W	22	$9(22-4) = 162$	6	G
D	3	$9(3-4) = -9$	17	R
E	4	$9(4-4) = 0$	0	A
X	23	$9(23-4) = 171$	15	P
Z	25	$9(25-4) = 189$	7	H
Y	24	$9(24-4) = 180$	24	Y

PT = CRYPTOGRAPHY

6.) Encrypt,

$$C = E(K_1 * P + K_2) \pmod{26} \quad K_1 = 13, K_2 = 4$$

PT		$K_1 * P$	$+ K_2$		
I	8	$13 \times 8 = 104$	$104 + 4 = 108$	4	E
N	13	$13 \times 13 = 169$	$169 + 4 = 173$	17	R
P	15	$13 \times 15 = 195$	$195 + 4 = 199$	17	R
V	20	$13 \times 20 = 260$	$260 + 4 = 264$	4	E
T	19	$13 \times 19 = 247$	$247 + 4 = 251$	17	R
A	0	$13 \times 0 = 0$	$0 + 4 = 4$	4	E
L	11	$13 \times 11 = 143$	$143 + 4 = 147$	17	R
T	19	$13 \times 19 = 247$	$247 + 4 = 251$	17	R
E	4	$13 \times 4 = 52$	$52 + 4 = 56$	4	E
R	17	$13 \times 17 = 221$	$221 + 4 = 225$	17	R

$K_1 = 13$ is not coprime with 26

The entire 26 letter alphabet is compressed into just two CT (E & R)

$$7. \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}^{-1} = (3 \times 7) - (5 \times 2) = 21 - 10 = 11$$

$$11^{-1} \text{ mod } 26 = 26 = 11(2) + 4$$

$$1 = 4 - 3(1)$$

$$11 = 4(2) + 3 \Rightarrow 1 = 4 - 3(11 - 4(2))$$

$$4 = 3(1) + 1$$

$$1 = (4) - (11)$$

$$1 = (26 - 11(2)) - (11) = -6(11) - (11) + 3 \times 26$$

$$1 = (-7)11 + 3 \times 26, \quad -7 + 26 = 19$$

$$11^{-1} \text{ mod } = 19$$

$$19 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} = \begin{bmatrix} 133 & -38 \\ -95 & 57 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

$$K^{-1} = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

$$8. \begin{bmatrix} 11 & 13 \\ 2 & 3 \end{bmatrix} = (11 \times 3) - (13 \times 2) = (33 - 26) = 7$$

$$7^{-1} \text{ mod } 26 \Rightarrow 26 = 7(3) + 5 \Rightarrow 1 = 5 - 2(2)$$

$$7 = 5(1) + 2$$

$$1 = 5 - 2(7 - 5)$$

$$5 = 2(2) + 1$$

$$1 = 3(5) - 2(7)$$

$$\Rightarrow 1 = 3(26 - 7(3)) - 2(7)$$

$$1 = (-1)2 + 3(26)$$

$$-1 + 26 = 15$$

$$7^{-1} \text{ mod } 26 = 15$$

$$15 \begin{bmatrix} 3 & -13 \\ -2 & 11 \end{bmatrix} = \begin{bmatrix} 45 & -195 \\ -30 & 165 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 19 & 13 \\ 22 & 9 \end{bmatrix}$$

$$9. \text{ Is the key valid? } \begin{bmatrix} 2 & 14 \\ 13 & 21 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 14 \\ 13 & 21 \end{bmatrix} = (2 \times 21) - (13 \times 14) = (21 - 182) = -160$$

160 does not co-prime with 26

10. "SOLVED"

3 8

7 11

for encryption $= (P \times K) \bmod 26$

$$\begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} = (3 \times 11) - (7 \times 8) = (33 - 56) = 23$$

23 is non-zero

23 is coprime with 26

plaintext :- SOLVED

$$(i) P = \begin{bmatrix} S \\ O \end{bmatrix} = \begin{bmatrix} 18 \\ 14 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} \begin{bmatrix} 18 \\ 14 \end{bmatrix} = \begin{bmatrix} 3 \times 18 + 8 \times 14 \\ 7 \times 18 + 11 \times 14 \end{bmatrix} = \begin{bmatrix} 166 \\ 280 \end{bmatrix} \bmod 26 = \begin{bmatrix} 10 \\ 20 \end{bmatrix} = \begin{bmatrix} K \\ W \end{bmatrix}$$

$$(ii) P = \begin{bmatrix} L \\ V \end{bmatrix} = \begin{bmatrix} 11 \\ 21 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} \begin{bmatrix} 11 \\ 21 \end{bmatrix} = \begin{bmatrix} 3 \times 11 + 8 \times 21 \\ 7 \times 11 + 11 \times 21 \end{bmatrix} = \begin{bmatrix} 201 \\ 308 \end{bmatrix} \bmod 26 = \begin{bmatrix} 19 \\ 22 \end{bmatrix} = \begin{bmatrix} T \\ W \end{bmatrix}$$

$$(iii) P = \begin{bmatrix} E \\ D \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \times 4 + 8 \times 3 \\ 7 \times 4 + 11 \times 3 \end{bmatrix} = \begin{bmatrix} 6 \\ 1 \end{bmatrix} \bmod 26 = \begin{bmatrix} 10 \\ 1 \end{bmatrix} = \begin{bmatrix} K \\ J \end{bmatrix}$$

CT = KUTWKJ